Faculty of Engineering

Department of Biomedical Engineering

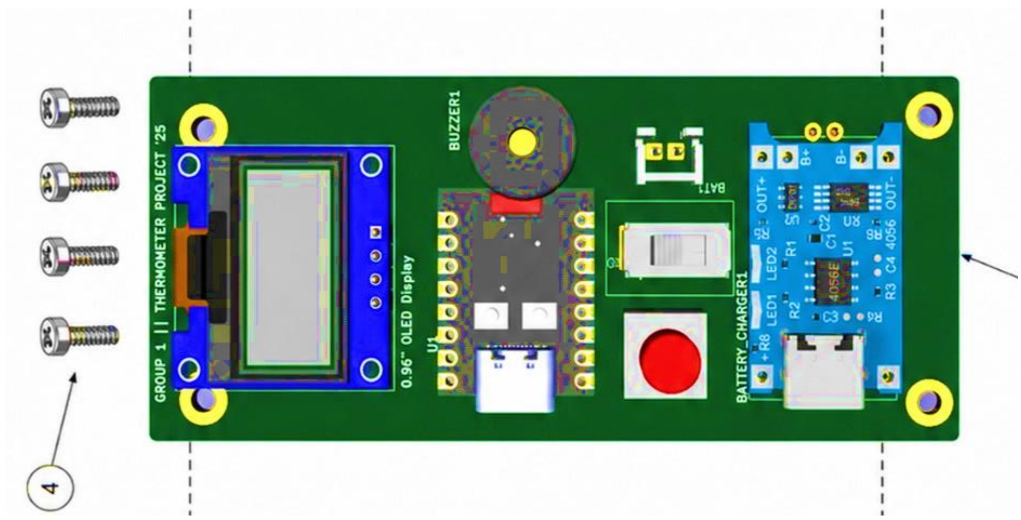
Non-Contact Type Clinical Thermometer Power

And Control Board Assembly

PART NUMBER:001-MIY-009

REV:1

DATE:17.05.2026



CHANGE RECORDS

FIRST RELEASE

CREATED BY			
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REVISIONS

Rev. No	Change Description	Date	QA Approval

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1. DRAWINGS

1.1 COMPONENTS LOCATIONS

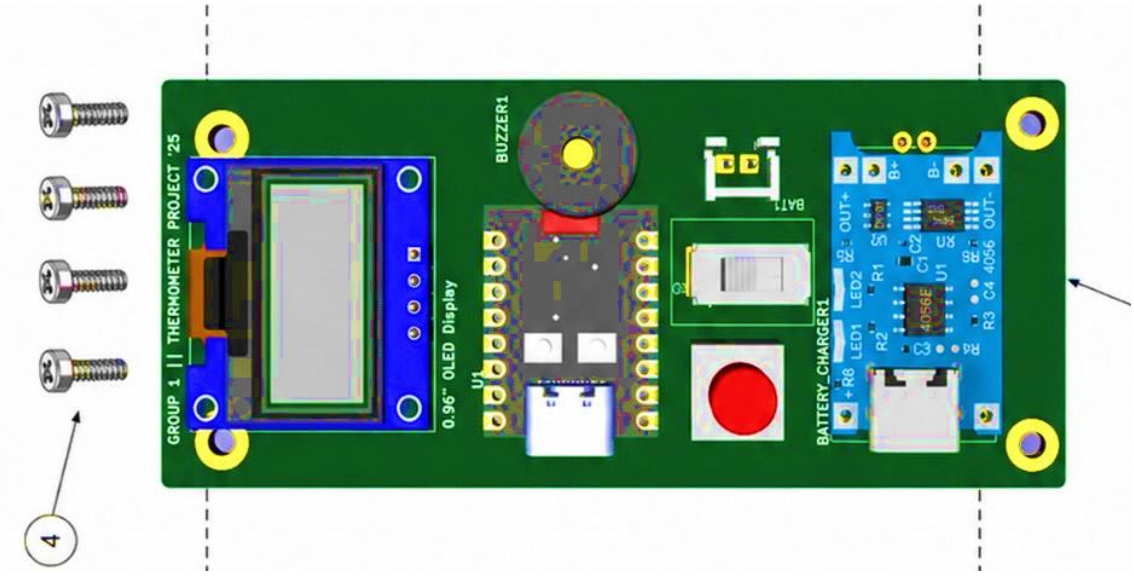

Figure1:PCB Model

1.2 ELECTRICAL INTERFACE

Item	P/N	Description	Location	Qty
1	KIT-1	ESP32-C3 Zero Mini MCU Board	Center	1
2	DISP-1	0.96" OLED Display Module	Left side	1
3	BZ-1	Buzzer	Top center	1
4	SW-1	Slide Power Switch	Center-right	1
5	PB-1	Push Button Switch	Lower center-right	1
6	KIT-3	TP4056 Battery Charger Module	Right side	1
7	BAT-1	Battery Connector / Input	Near TP4056 module	1
8	PCB-1	Main Printed Circuit Board	Base board	1
9	H1	Mounting Hole	Top-left corner	1
10	H2	Mounting Hole	Bottom-left corner	1
11	H3	Mounting Hole	Top-right corner	1
12	H4	Mounting Hole	Bottom-right corner	1
13	SCR-1	Mounting Screws	Left side assembly	4

1.3 CIRCUIT DIAGRAM

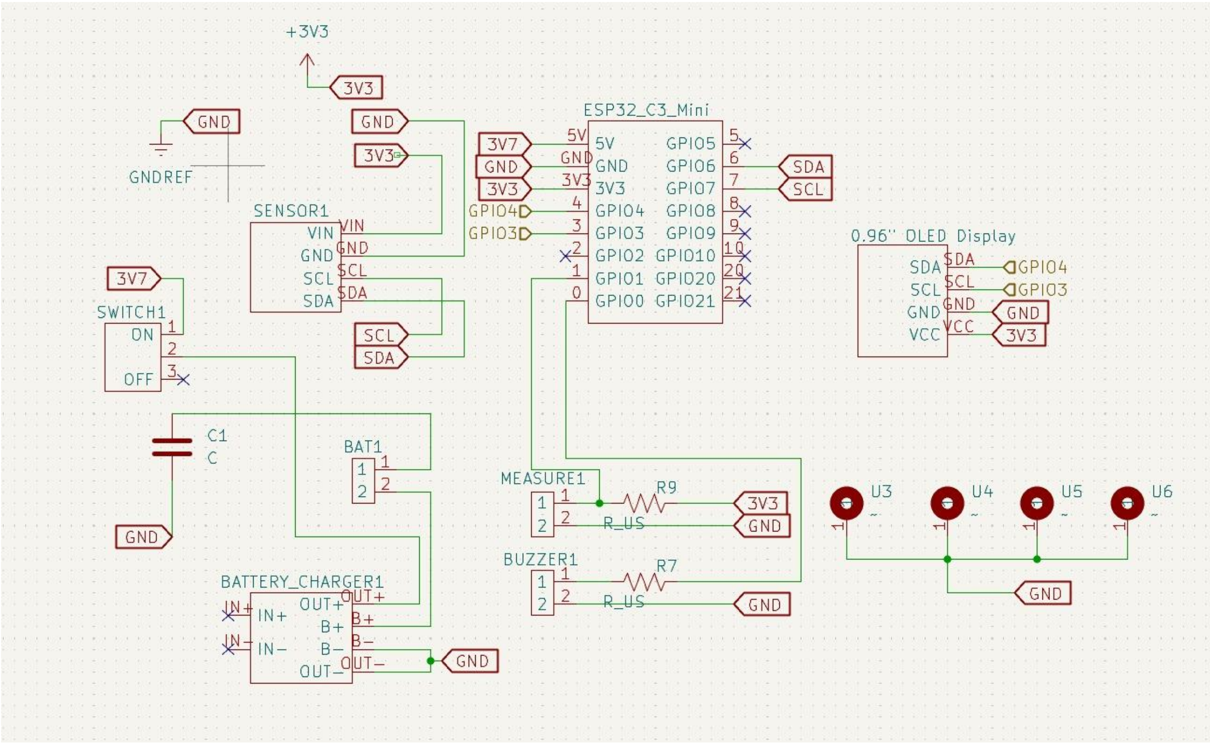

Figure 2: Circuit Diagram Schematic

2. PART LIST (BILL OF MATERIAL-BOM)

Item	P/N	Description	Location	Quantity
1	01-PRT-001	MCU Kit, ESP32 C3 Mini	ESP32_C3_Mini	1
2	01-PRT-002	MLX90614, Infrared Camera Sensor	SENSOR1	1
3	01-PRT-003	TP4056, Battery Charger Module	BATTERY_CHARGER1	1
4	01-PRT-004	Display Kit, 0.96" LCD Display	0.96" OLED Display	1
5	01-PRT-005	On /Off Slider Switch	SWITCH1	1
6	01-PRT-006	Measurement Button	MEASURE1	1
7	01-PRT-007	Buzzer	BUZZER1	1

Item	P/N	Description	Location	Quantity
8	01-PRT-008	Resistor	R7, R9	2
9	01-PRT-009	JST Connector	BAT1	1
10	01-PRT-010	Lithium Ion Battery	Connects to BAT1	1
11	01-PRT-011	Sensor PCB	System Assembly	1
12	01-PRT-012	Housing Assembly	System Assembly	1

3. TECHNICAL DESCRIPTIONS

3.1. Soldering shall be done according to the latest revision of IPC-J-STD-001 and inspected per IPC-A-610. Clean the solder joints with Isopropyl Alcohol and a brush before inspection.

3.2. The first component to be installed is the Battery Charger Module (`BATTERY_CHARGER1` / TP4056) and the battery connector (`BAT1`). After installation, connect the rechargeable Li-Po battery to `BAT1` and provide 5V power via the USB-C input on the charger module. Measure the voltage across the `OUT+` and `OUT-` terminals to verify battery output. It should be between 3.7 VDC and 4.2 VDC. Remove the battery and USB power connections.

3.3. Solder the remaining surface mount components (resistors `R7`, `R9`, capacitor `C1`), the switch (`SWITCH1`), the measurement button (`MEASURE1`), the buzzer (`BUZZER1`), the infrared sensor (`SENSOR1`), and the MCU Kit (`ESP32_C3_Mini`). Perform a continuity check to ensure there are no short circuits or open circuits, paying special attention to the `3V3` and `GND` lines.

3.4. Insert the battery and toggle `SWITCH1` to the ON position to verify system power. Measure the voltage at the `3V3` pin on the ESP32-C3 Mini (relative to `GND`). It should be 3.3V ± 0.2V.

3.5. Solder the 0.96" OLED Display. Power the system on again and verify that the display illuminates properly without dropping the `3V3` line voltage.

3.6. Connect the ESP32 C3 Mini directly to a computer via its integrated USB port to upload the firmware. Once the firmware is flashed, disconnect the device from the PC; the device operates entirely on battery power and requires no PC data acquisition or external DC power supplies for the functional testing phases.

3.7. Trigger the `MEASURE1` button and observe the non-contact temperature readings for ambient and object temperatures on the OLED display. Verify that the visual interface responds accurately according to the programmed OLED indicators, ensuring it operates correctly without displaying arbitrary strings such as "BELOW THE RANGE" or "ABOVE THE RANGE".